a)

The true trajectory is presented in *Figure 1*.

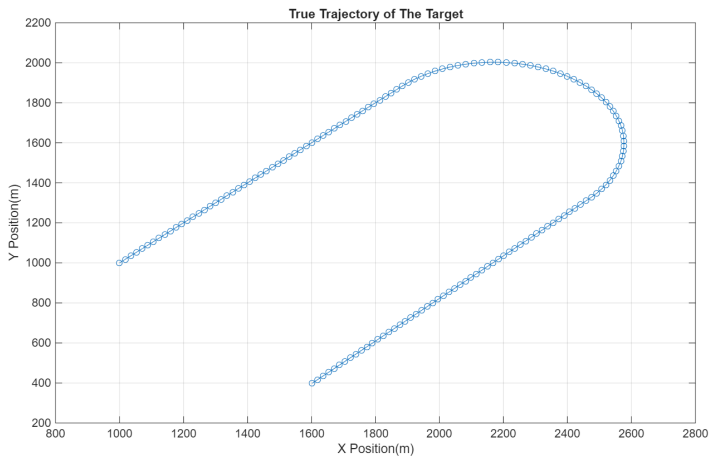

Figure 1: True Trajectory

b)

The noisy measurements are also included on top of true trajectory in *Figure 2*.

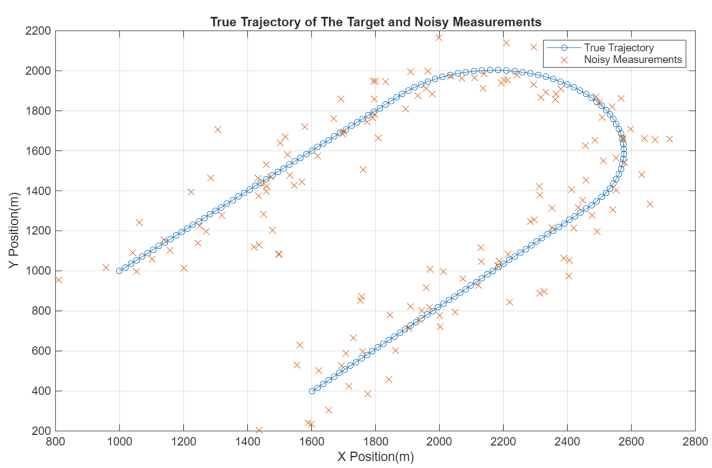

Figure 2: True Trajectory and Noisy Measurements

c)

Observed estimation and predictions are given in *Figure 3*:

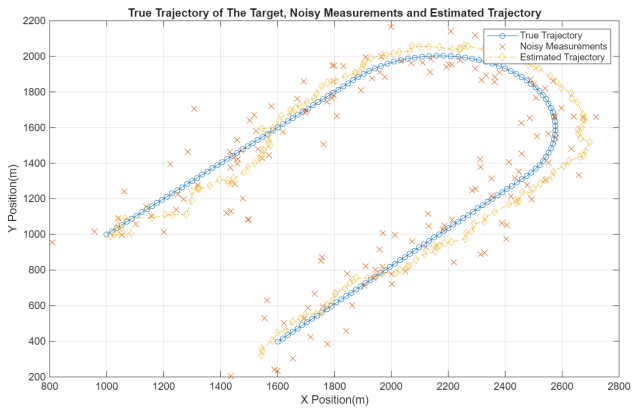

Figure 3: Estimated Trajectory of The Target

Q:Which one of the errors $e_{k|k}$ or $e_{k|k-1}$ is larger most of the times? Why?

Ans: The prediction error $e_{k|k-1}$ is larger. The reason is that the estimation error $e_{k|k}$ correspond to the error of the state that is corrected by the measurements. Which gives more information about the actual state. Estimation and the prediction errors are given in *Figure 4*:

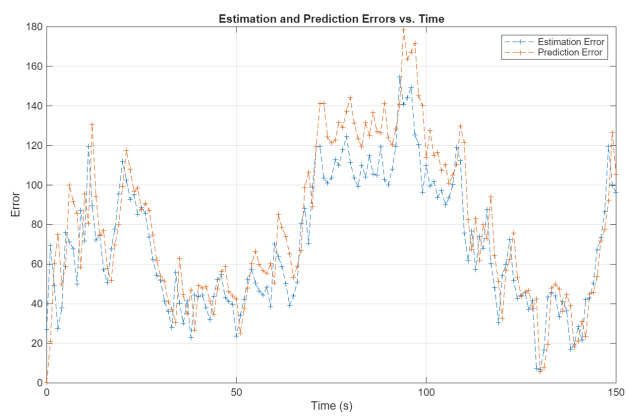

Figure 4: Estimation and Prediction Errors

The corresponding RMS errors are calculated as the following:

- $RMS_{k|k-1} = 88.1420$
- $RMS_{k|k} = 77.7896$

d)

Case 1: Process noise is fixed. The measurement noise is increased and decreased 100 times. The resulting estimated target trajectories are presented in *Figure 5* for 100 times increase and *Figure 6* for 100 times decrease.

Figure 5: 100x Increased Measurement Noise, Process Noise Fixed

The significant change in measurement noise tells our estimator how reliable are our measurements. If it is increased 100 times, we expect the estimator to rely on the measurement model more; Thus, it would follow the measurement updates more slowly and rely on model of the system more. This behavior can be observed in Kalman measurement update equations.

$$S_k = CP_{k|k-1}C^T + R$$

$$K_k = P_{k|k-1}C^T S^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - \hat{y}_{k|k-1})$$

In the first equation, we can say that the innovation covariance resembles a combination of measurement and state covariances. It can be interpreted as the total deviation we expect from the true trajectory.

In the second equation, the Kalman gain calculation can be interpreted as the ratio of the state's covariance with the innovation covariance, which is the combination of state and measurement covariances. Thus, Kalman gain can be interpreted as the ratio of the expected deviation of state to the total expected deviation from the true trajectory.

In the third equation, we see that the Kalman gain multiplies the difference between the measurement and the predicted measurement, and adds this to the predicted state to combine our prediction with the measurement results. If the measurement noise increases, Kalman gain would decrease. Thus, the residual $(y_k - \hat{y}_{k|k-1})$ component in the estimate , i.e., the contribution of measurement, would decrease.

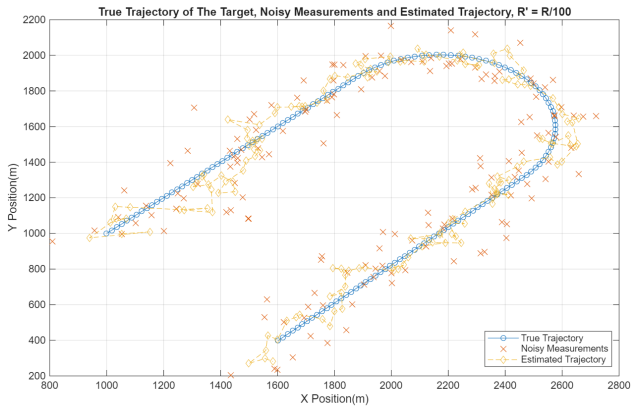

Figure 6: 100x Decreased Measurement Noise, Process Noise Fixed

It's the same logic but inverse for the 100 times decrease. It increases the trust in measurements and our estimate would have more contribution from measured state.

Case 2: Measurement noise is fixed. The process noise is increased and decreased 100 times. The resulting estimated target trajectories are presented in *Figure 7* for 100 times increase and *Figure 8* for 100 times decrease.

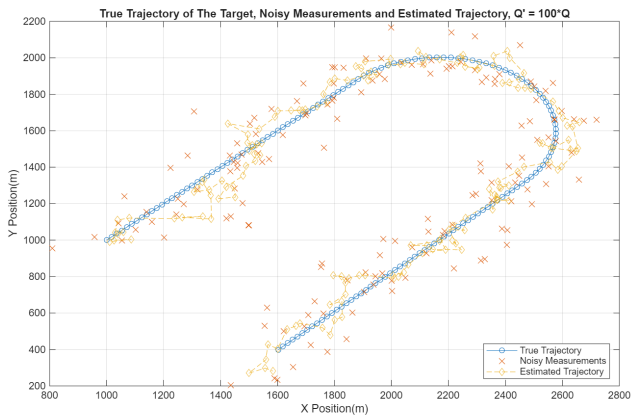

Figure 7: 100x Increased Process Noise, Measurement Noise Fixed

The significant change in the process noise tells our estimator how reliable our motion model is. If it is increased 100 times, we expect high modeling errors and tell the estimator to account for it. If it is decreased 100 times, we assume that the mathematical model of the system is accurate.

By the same logic in the fixed measurement noise case, we expect the Kalman filter to determine in which ratio the system model and the measurements would be incorporated in the estimation. The effect of the process noise can be observed from the covariance prediction equation:

$$P_{k|k-1} = AP_{k-1|k-1}A^T + BQB^T$$

If Q is increased, predicted state covariance also increases. Thus, the Kalman gain (Which is proportional with $\frac{P}{P+R}$), is increased, and the measurements are incorporated more in the estimated state.

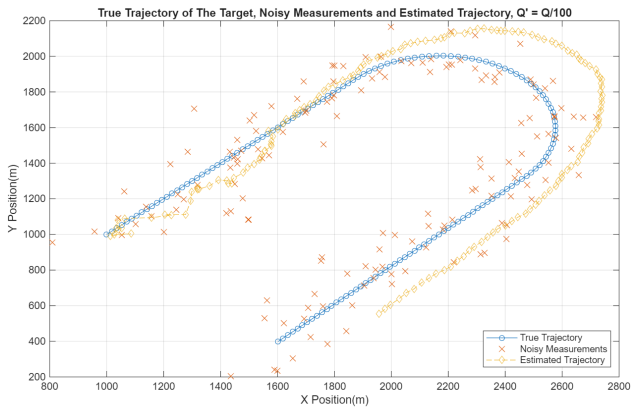

Figure 8: 100x Decreased Process Noise, Measurement Noise Fixed

If Q is decreased, Kalman gain is also decreased and the motion model is incorporated more in the estimated state.

The effects of increasing process noise and decreasing measurement noise are nearly the same. We can also say this for decreasing process noise and increasing measurement noise, as the change in the Kalman gain determines the outcome.

e)

Nothing to talk about, polar measurements are generated.

f)

The EKF and UKF filters are used for estimation of the state with the same polar measurements. The resulting estimation and prediction errors are presented in *Figure 9*

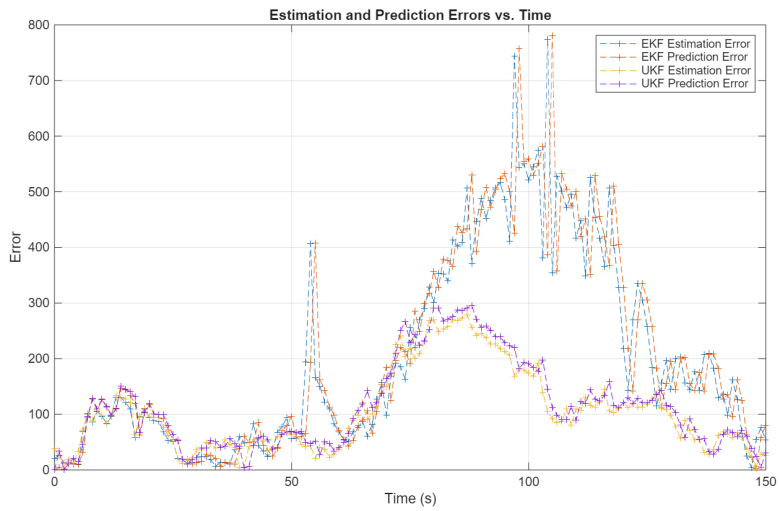

Figure 9: EKF and UKF, Estimation and Prediction Errors With Respect to Time

The resulting RMS errors for the filters are calculated as the following:

- EKF $RMS_{k|k-1} = 276.8208$
- UKF $RMS_{k|k-1} = 138.3648$
- EKF $RMS_{k|k} = 269.2670$
- UKF $RMS_{k|k} = 127.6732$

As expected, UKF performs better on high non-linearity. This is due to the difference in linearization methods.

In EKF, the linearization is directly applied by first-order Taylor Series approximation. If the measurement frequency is high enough, or the non-linearities are small that the first-order approximation still holds with some accuracy, EKF is a good analytical solution to this problem. However, when the nonlinearities increase, EKF's assumption fails and increases errors.

UKF on the other hand, does not include a linearization dependent on the analytical expression of the non-linearity. It calculates sigma points prom the prior, calculates the outcome of the nonlinear function and then reconstructs the posterior. It handles highly nonlinear cases better, such as polar measurements.

In the result, both of these methods seem to perform worse than the plain Kalman filter approach when we consider the errors. But this is not true, as the measurement error in the polar measurements will be larger when the target

gets further away from the origin (since the measurement model in this scope assumes that the sensor is at the origin), the error that corresponds to x and y increases, due to angular error.